



CSC 6011 - Theory of Computation - Lecture 8: Undecidability

Time: Mon & Wed, 4:00 - 5:20 PM

Location: Teaching A, 107

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Lecture 8

Last time:

- Decision problems for automata and grammars

A_{DFA} , A_{NFA} , E_{DFA} , EQ_{DFA} , A_{CFG} , E_{CFG} are decidable

A_{TM} is T-recognizable

Today:

- A_{TM} is undecidable
- The diagonalization method
- $\overline{A_{\text{TM}}}$ is T-unrecognizable
- Other undecidable languages/problems (e.g., Halting)

Recall: Acceptance Problem for TMs

Let $A_{\text{TM}} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$

Today's Theorem: A_{TM} is not decidable

Proof uses the diagonalization method, which we will introduce first.

The Size of Infinity

How to compare the relative sizes of infinite sets?

Georg Cantor (~1890s) had the following idea.

Defn: Say that sets A and B have the same size if there is a “one-to-one” and “onto” function $f: A \rightarrow B$


 $x \neq y$
 $\rightarrow f(x) \neq f(y)$

“injective”


Range of $f = B$

“surjective”

Note: this “same-size” definition works for infinite sets too.

The “size” of an infinite set is also called cardinality

We call such an f a bijection (or one-to-one correspondence).

Countable Sets

Defn: A set is countable if it is finite or it has the same size as $\mathbb{N} = \{1, 2, 3, \dots\}$
(is countably infinite).

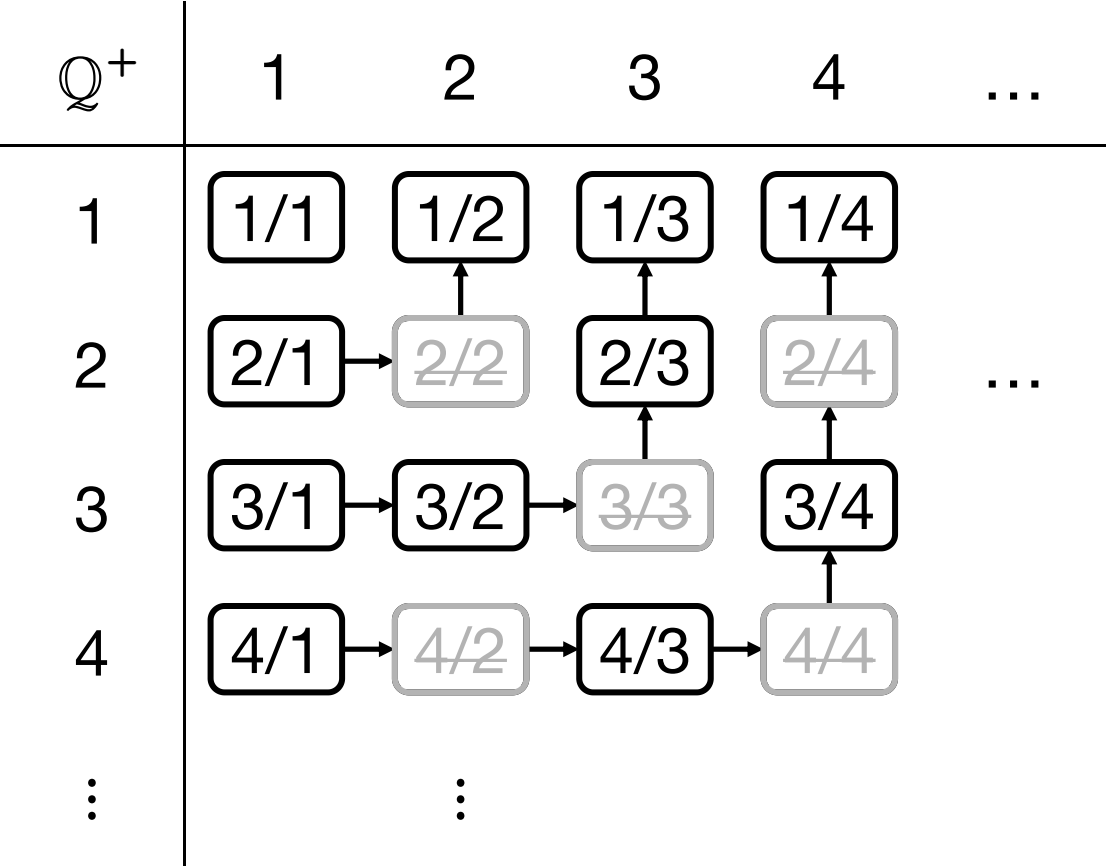
Claim: Both $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Show $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countable:

		n	$f(n)$		
		1	0		
		2	-1		
$\mathbb{N}$		3	1		$\mathbb{Z}$
		4	-2		
		5	2		
		6	-3		
		7	3		
		$\vdots$	$\vdots$		

Countable Sets

How to show $\mathbb{Q}^+ = \{ m/n \mid m, n \in \mathbb{N} \}$ is countable?



	<i>n</i>	<i>f</i> (<i>n</i>)	
$\mathbb{N}$	1	1/1	$\mathbb{Q}^+$
	2	2/1	
	3	1/2	
	4	3/1	
	5	3/2	
	6	2/3	
	7	1/3	
	8	4/1	
	9	4/3	
	10	3/4	
	11	1/4	
	...	...	

$\mathbb{R}$ is Uncountable – Diagonalization

Let $\mathbb{R}$ = all real numbers

Theorem: $\mathbb{R}$ is uncountable

Proof by contradiction via diagonalization:

Assume $\mathbb{R}$ is countable

So there is a bijection $f: \mathbb{N} \rightarrow \mathbb{R}$

n	$f(n)$
1	2.718281828...
2	3.141592653...
3	0.000000000...
4	1.414213562...
5	0.142857242...
6	0.207879576...
7	1.234567890...
$\vdots$	$\vdots$

$\mathbb{R}$ is Uncountable – Diagonalization

n	$f(n)$
1	2. 7 18281828...
2	3.1 4 1592653...
3	0.00 0 000000...
4	1.414 2 13562...
5	0.1428 5 7242...
6	0.20787 9 576...
7	1.234567 8 90...
$\vdots$	$\vdots$

“Diagonalization”

Demonstrate a number $x \in \mathbb{R}$ that is not in the list:

Take the diagonal line, change each digit:

$$x = 0.\underset{\neq 7}{8}\underset{\neq 4}{5}\underset{\neq 0}{1}\underset{\neq 2}{6}\underset{\neq 5}{1}\underset{\neq 9}{8}\underset{\neq 8}{2}\dots$$

differs from the n^{th} number in the n^{th} digit
so cannot be the n^{th} number for any n .

Hence x is not in the list.

Therefore f is not surjective, so it cannot be a bijection.

$\mathbb{R}$ is Uncountable – Corollaries

Let $\mathbb{L} =$ all languages $= \mathcal{P}(\Sigma^*)$

Corollary 1: $\mathbb{L}$ is uncountable

Proof: There is a one-to-one correspondence between $\mathbb{L}$ and $\mathbb{R}$.

Σ^*	{ ε , 0, 1, 00, 01, 10, 11, 000, ... }
$A \in \mathbb{L}$	{ 0, 00, 01, ... }
$f(A)$	0 1 0 1 1 0 0 0 ...

Observation: $\Sigma^* = \{\varepsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$ is countable.

Let $\mathbb{M} =$ all Turing machines

Observation: $\mathbb{M}$ is countable.

Because $\{ \langle M \rangle \mid M \text{ is a TM} \} \subseteq \Sigma^*$.

Corollary 2: Some language is not decidable.

Because there are more languages than TMs.

We will show a specific language A_{TM} is not decidable.

$\mathbb{R}$ is Uncountable – Corollaries

Let $\mathcal{L}$ = all languages = $\mathcal{P}(\Sigma^*)$

Corollary 1: $\mathcal{L}$ is uncountable

Proof: There is a one-to-one correspondence between $\mathcal{L}$ and $\mathbb{R}$.

Σ^*	{ ϵ , 0, 1, 00, 01, 10, 11, 000, ...
$A \in \mathcal{L}$	{ 0, 00, 01, ...
$f(A)$	0 1 0 1 1 0 0 0 ...

Observation: $\Sigma^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}$

Let $\mathcal{M}$ = all Turing machines

Observation: $\mathcal{M}$ is countable

Because $\{ \langle M \rangle \mid M \text{ is a TM} \} \subseteq \Sigma^*$

Corollary 2: Some language is not recursive

Because there are more languages than Turing machines

We will show a specific language is not recursive

Mini-Quiz 8.1

Hilbert's 1st question (Continuum Hypothesis):

Is there a set of intermediate cardinality between $\mathbb{N}$ and $\mathbb{R}$?

(a) Yes.

(b) No.

(c) I don't know.

Gödel and Cohen proved that CH cannot be proved or disproved using standard axioms of mathematics (ZFC).

A_{TM} is undecidable

Recall $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$

Theorem: A_{TM} is not decidable

Proof by contradiction: Assume some TM H decides A_{TM} .

So H on $\langle M, w \rangle = \begin{cases} \text{Accept} & \text{if } M \text{ accepts } w \\ \text{Reject} & \text{if not} \end{cases}$

Use H to construct TM D

$D =$ “On input $\langle M \rangle$

1. Simulate H on input $\langle M, \langle M \rangle \rangle$
2. *Accept* if H rejects. *Reject* if H accepts.”

D accepts $\langle M \rangle$ iff M doesn't accept $\langle M \rangle$.

D accepts $\langle D \rangle$ iff D doesn't accept $\langle D \rangle$.

Contradiction.

Why is this proof a diagonalization?

		All TM descriptions					
		$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	...	$\langle D \rangle$
All TMs	M_1	acc	rej	acc	acc	...	
	M_2	rej	rej	rej	rej		
	M_3	acc	acc	acc	acc	...	
	M_4	rej	rej	acc	acc		
	$\vdots$			$\vdots$			
	D	rej	acc	rej	rej		?

D accepts $\langle M \rangle$ iff M doesn't accept $\langle M \rangle$.

D accepts $\langle D \rangle$ iff D doesn't accept $\langle D \rangle$.

Contradiction.

$\overline{A_{TM}}$ is T-unrecognizable

Lemma: If A and $\overline{A}$ are T-recognizable, then A is decidable

Proof: Let TM M_1 and M_2 recognize A and $\overline{A}$.

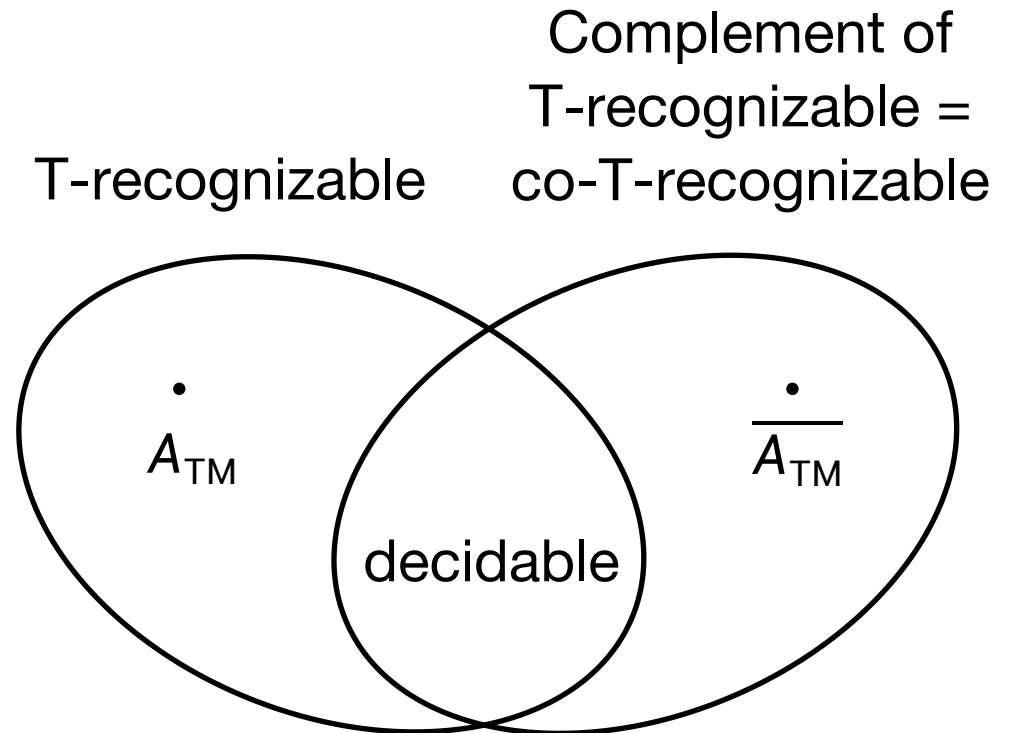
Construct TM T deciding A .

T = “On input w

1. Run M_1 and M_2 on w in parallel until one accepts.
2. If M_1 accepts then *accept*.
If M_2 accepts then *reject*.”

Corollary: $\overline{A_{TM}}$ is T-unrecognizable

Proof: A_{TM} is T-recognizable but undecidable



Mini-Quiz 8.2

From what we've learned, which closure properties can we prove for the class of T-recognizable languages?

Choose all that apply:

- (a) Closed under union.
- (b) Closed under intersection.
- (c) Closed under complement.
- (d) Closed under concatenation.
- (e) Closed under star.

The Halting Problem is undecidable

Let $\text{HALT}_{\text{TM}} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on } w \}$.

Theorem: HALT_{TM} is undecidable.

Proof by contradiction: Assume TM H decides HALT_{TM} .
 H always halts and answers whether M halts on input w .

Use H to construct TM D :

$D =$ “On input $\langle M \rangle$:

1. Run H on input $\langle M, \langle M \rangle \rangle$.
2. If H answers Yes, **loop forever**.
3. If H answers No, **halt**.”

D halts on $\langle M \rangle$ iff M does not halt on $\langle M \rangle$.

D halts on $\langle D \rangle$ iff D does not halt on $\langle D \rangle$.

Contradiction.

Why is this proof a diagonalization?

		All TM descriptions					
		$\langle M_1 \rangle$	$\langle M_2 \rangle$	$\langle M_3 \rangle$	$\langle M_4 \rangle$	...	$\langle D \rangle$
All TMs	M_1	1	0	1	1	...	
	M_2	0	0	0	0		
Entry a_{ij} : 1 = halts 0 = loops	M_3	1	1	1	1	...	
	M_4	0	0	1	1		
	$\vdots$			$\vdots$			
$D = M_k$		0	1	0	0		?

D halts on $\langle M \rangle$ iff M does not halt on $\langle M \rangle$.
 D halts on $\langle D \rangle$ iff D does not halt on $\langle D \rangle$.
 Contradiction.

$$a_{kk} = 1 - a_{kk}$$

Quick review of today

1. Showed that $\mathbb{N}$ and $\mathbb{R}$ have different cardinalities (sizes) to introduce the Diagonalization Method.
2. A_{TM} is undecidable (using Diagonalization).
3. $\overline{A_{TM}}$ is T-unrecognizable (using the fact that if A and $\overline{A}$ are T-recognizable, then A is decidable).
4. Show that $HALT_{TM}$ is undecidable (using Diagonalization).

These slides are adapted from Michael Sipser, *18.404J Theory of Computation*, Fall 2020. Massachusetts Institute of Technology:

- Source: MIT OpenCourseWare <https://ocw.mit.edu>

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